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SUPERSEDING
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2 April 1981

MILITARY SPECIFICATION

FORWARD AREA REFUELING EQUIPMENT

(FARE)

This specification is approved for use within the USA Belvoir Research, Development, and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable fuel-dispensing system for use in forward areas to refuel aircraft, ground vehicles, and other powered equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin.
P-D-680	- Dry Cleaning Solvent.
NN-P-530	- Plywood, Flat Panel.
QQ-S-781	- Strapping, Steel, and Seals.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930
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PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-C-843	- Cushioning Material, Cellulosic.
PPP-P-40	- Packaging and Packing of Hand Tools.
PPP-T-60	- Tape: Packaging, Waterproof.
PPP-T-97	- Tape, Pressure-Sensitive Adhesive, Filament Reinforced.

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MIL-P-116	- Preservation, Methods of.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-G-3056	- Gasoline, Automotive, Combat.
MIL-T-5624	- Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant.
MIL-F-46162	- Fuel, Diesel, Referee Grade.
MIL-C-46168	- Coating Aliphatic Polyurethane, Chemical Agent Resistant.
MIL-F-52556	- Filter-Separator, Liquid Fuel: Frame Mounted, 100 GPM Capacity.
MIL-P-52746	- Pump Assembly, Flammable Liquid, Centrifugal, Self-Priming, 2-Inch, 100 GPM at 100 Feet Total Dynamic Head.
MIL-N-52747	- Nozzle Assembly, Closed-Circuit Refueling.
MIL-N-52748	- Nozzle, Adapter, Closed-Circuit to Gravity Fill.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MS35338	- Washer, Lock-Spring, Helical, Regular (Medium) Series, (in/mm).
MS35649	- Nut, Plain-Hexagon, Machine Screw UNC-2B.

HANDBOOKS

MILITARY

MIL-HDBK-113	- Guide for the Selection of Lubricants, Fluids, Preservatives and Speciality Products for use in Ground Equipment Systems.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. DEPARTMENT OF COMMERCE

Product Standard

PS-1 - Softwood, Plywood; Construction and Industrial.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

DRAWINGS

ME

TA13219E0500 - Forward Area Refueling Equipment.

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

(Application for copies should be addressed to the American Society of Mechanical Engineers Inc., 345 East 47th Street, New York, NY 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

AMERICAN WELDING SOCIETY (AWS)

B2.1 - Standard for Welding Procedure and Performance Qualification.

(Application for copies should be addressed to the American Welding Society, Inc., 2501 Northwest Seventh Street, Miami, FL 33125.)

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ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Department of Defense Materiel On Open Top Cars.

(Application for copies should be addressed to the Association of American Railroads, 1920 L Street, NW, Washington, DC 20036.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/HPMA-HP 1983 American National Standard for Hardwood and Decorative Plywood.

(Application for copies should be addressed to American National Standards Institute, 1430 Broadway, New York, NY 10018.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The forward area refueling equipment, hereinafter referred to as FARE, shall be as shown on TA13219E0500 and as specified herein. The major components of FARE include the pump-engine assembly, two discharge hose and fitting kits, and two suction hose and ground rod kits.

3.1.1 Drawings. The drawings forming a part of this specification are end product drawings. No deviation from the prescribed dimensions or tolerances is permissible without prior approval of the contracting officer. Where tolerances could cumulatively result in incorrect fits, the contractor shall provide tolerances within those prescribed on the drawings to insure correct fit, assembly, and operation of the FARE. Any data (e.g. shop drawings, layouts, flow sheets, processing procedures, etc.) prepared by the contractor or obtained from a vendor to support fabrication and manufacture of the production item shall be made available, upon request, for inspection by the contracting officer or designated representative.

3.2 First article. Unless otherwise specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of the FARE system from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish FARE systems conforming to this specification.

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3.3 Material. Material shall be as specified herein and on the applicable drawings. Material not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Materiel deterioration prevention and control. The FARE system shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operation and storage environment to which the FARE may be exposed.

3.3.2 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.3 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.4 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the FARE system may be newly fabricated from recovered materials to the maximum extent practicable, provided the FARE systems produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the FARE systems.

3.4 Threads and threaded connections. All pipe and machine screw threads shall conform to FED-STD-H28. Prior to assembly, threads on all screwed fittings shall be coated with sealing compound conforming to MIL-S-7916.

3.5 Components. The FARE when assembled for field movement shall have all components as specified herein securely contained within the frame assemblies as shown in TA13219E0500.

3.6 Pump. The pump shall be in accordance with MIL-P-52746.

3.7 Filter-separator. The filter-separator assembly shall be in accordance with MIL-F-52556.

3.8 Nozzle. The nozzle shall be in accordance with MIL-N-52747.

3.9 Nozzle adapter. The nozzle adapter shall be in accordance with MIL-N-52748.

3.10 Environmental.

3.10.1 Operating temperature. The FARE shall perform as specified herein in any ambient temperature from +125 °F to -25 °F.

3.10.2 Storage temperature. The FARE shall not be damaged by storage in any ambient temperature from +160 °F to -50 °F.

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3.11 Reliability. The specified mean-time-between-failure (MTBF) shall be 150 hours when the FARE is tested as specified in 4.6.1.3.

3.12 Maintenance ratio. The FARE shall have a maintenance ratio not greater than 0.08. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.13 Performance.

3.13.1 Capacity. The FARE when assembled as specified herein for operation, and tested as specified in 4.6, shall deliver not less than 50 gallons per minute (gpm) simultaneously through each of two nozzles specified in 3.8 with nozzle adapters specified in 3.9.

3.13.2 Compatability. The FARE components shall be compatible when connected and shall not leak when tested as specified in 4.6.1.5.2.

3.13.3 Continuity. The static electric ground assemblies for the pump, filter-separator and nozzles shall show a continuous electrical circuit when tested as specified in 4.6.1.5.3.

3.14 Lubricants. The procedure for the selection of lubricants shall be in accordance with section 5 of MIL-STD-838. Lubricants selected shall be in accordance with chapter 2 of MIL-HDBK-113. When the specification of the lubricant selected includes a requirement for a qualified products list (QPL) the lubricant supplied shall be from a source that is listed on the applicable QPL.

3.15 Treatment and painting. The portions of the FARE normally painted internally and externally shall be cleaned, treated, and painted in accordance with MIL-T-704, type F or G as applicable. Unless otherwise specified (see 6.2) top coat color shall be camouflage 383 conforming to MIL-C-46168.

3.16 Transportability. The FARE shall be suitable for transport by rail, highway, and air. The items shall meet the test requirements of Association of American Railroads (AAR) recommended procedures for conducting field impact tests of loaded freight cars.

3.17 Instructions. An instruction plate/placard shall be provided with a diagram showing the assembly of the FARE components and a brief description of the assembly procedure. The instruction plate/placard shall be attached in a conspicuous location.

3.18 Workmanship.

3.18.1 Castings and forgings. All parts, components, and assemblies of the FARE which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce strength or affect function.

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3.18.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause damage to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.18.3 Aluminum welding. The aluminum surfaces of parts to be welded shall be free from scale, paint, grease, oxide film, and other foreign matter. The welding shall be done by the inert gas, shielded-arc method. Welds may be ground, filed, buffed, or chipped but shall not be hammered.

3.18.4 Welding. The surfaces of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, other foreign matter, and shall be free from discontinuities such as base metal cracks adjacent to or behind welds, cracks in weld metal, undercutting of base metal, lack of fusion with parent metal or between parts, lack of penetration, lack of prescribed fit, or spatter. Parent materials, weld filler metals and fabrication techniques shall be as required to enable the FARE to conform to the examination and test requirements specified in section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the FARE shall be at the option of the contractor.

3.18.5 Welders. Before assigning any welder or welding operator to welding work covered by this specification, the contractor shall obtain certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

AWS B2.1, Standard for Welding Procedure and Performance Qualifications.
ASME Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

Contractors who make only horizontal welds need not qualify welds for "all position welding." The certification shall be made available for review by the contracting officer or designated representative upon request. The Government reserves the right to require the recertification of any welder or welding operator.

3.18.6 Bolted connections. Boltholes shall be accurately punched or drilled and shall have the burrs removed. Boltholes in structural members subject to variable loads shall be punched or drilled. Washers or lockwashers shall be provided where necessary. All bolts, nuts, and screws shall be correctly torqued and shall have full thread engagement. Matching thread areas securing bolts or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt.

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3.18.7 Riveting. Rivet holes in the sizes recommended by standard practice shall be accurately punched or drilled, and shall have all burrs removed. Rivets shall be neatly set, concentric with rivet holes, and in full contact with the surfaces of the members.

3.19 Safety. The FARE shall present no uncontrolled hazards to the user.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use any facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Acceptability criteria. FARE systems which conform to all requirements in sections 3 and 5 and pass all applicable examinations and tests in section 4 of this document will be considered acceptable by the Government.

4.1.3 Parts and components. Parts and components detailed on the drawings shall be inspected in accordance with the quality assurance provisions (QAP) shown on the drawings. The drawings specify the characteristics requiring QAP inspection, the sampling plan, and the basis for acceptance and rejection (see 6.7).

4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.3 and 4.8.1).
b. Quality conformance inspection (see 4.4).
c. Inspection comparision (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article inspection. First article inspection shall be performed on one complete unit when a first article sample is required (see 3.2 and 6.2). This inspection shall include the examination of 4.5 and tests of 4.6.1.

4.4 Quality conformance inspection. Quality conformance inspection shall include the examination of 4.5 and tests of 4.6.1.

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4.5 Examination. Each FARE system shall be examimed as specified herein for the following defects. Presence of one or more defects shall be cause for rejection.

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101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be corrosion resistant for the applicable storage and operating environment.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes, or treatments.
105. Used, rebuilt, or remanufactured components, pieces or parts incorporated in the FARE system.
106. Any part (or component) not in accordance with the QAP requirements as shown on the drawings.
107. Any dimension, other than those identified on the QAP requirements shown on the drawings, not as specified.
108. Threads and threaded assemblies not as specified.
109. Components missing or not as specified.
110. Hardware not as specified.
111. Identification marking, incomplete, or illegible.
112. Treatment and painting not as specified.
113. Paint color not as specified.
114. Workmanship not as specified.
115. Welding not as specified.
116. Welder/welding operator certification not as specified.
117. Instruction plate/placard not as specified.

4.6 First article tests. When a first article test is required, the first article shall be subjected to the tests specified in 4.6.1.

4.6.1 Tests.

4.6.1.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The FARE shall operate as specified herein without maintenance other than the contractor's recommended normal scheduled maintenance as established by the maintenance schedule prepared and submitted by the contractor prior to test. Before performing the following tests, the FARE shall be serviced with military fuels and lubricants designated for use at the ambient temperature at which the test is being conducted. The test fluid shall be gasoline conforming to MIL-G-3056, jet fuel conforming to MIL-T-5624, diesel fuel conforming to MIL-F-46162, or solvent conforming to P-D-680.

4.6.1.2 Environmental.

4.6.1.2.1 High temperature. The FARE shall be tested as specified in MIL-STD-810, method 501, procedure I. The maximum ambient storage temperature shall be 160 °F, and the maximum ambient operating temperature shall be 125 °F.

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The operating period of procedure I shall be 2 hours. The FARE shall be operated at rated capacity. Nonconformance to 3.11 shall constitute failure of this test.

4.6.1.2.2 Low temperature. The FARE shall be tested as specified in MIL-STD-810, method 502, procedure I. The minimum ambient storage temperature shall be -50 °F. The minimum ambient operating temperature shall be -25 °F. The operating period of procedure I shall be 2 hours. The FARE shall be operated at rated capacity. Nonconformance to 3.11 shall constitute failure of this test.

4.6.1.3 Reliability. The first article FARE shall be tested as specified in 4.6.1.5 with "accept" and "reject" criteria in accordance with figure 1, using the MTBF specified in 3.11. A failure is defined as any malfunction which the operator cannot correct within 30 minutes by adjustment, repair, or replacement action; using controls, on-equipment tools, or parts; and which causes or may cause:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Serious damage to the FARE system by continued operation.
c. Serious personnel safety hazards.

Nonconformance to 3.11 or 3.12 shall constitute failure of this test.

4.6.1.4 Maintainability.

4.6.1.4.1 Maintenance support evaluation. Maintenance operations required during initial production testing shall be accomplished to determine conformance to 3.12. Failure of the support elements to maintain the FARE as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.1.4.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.12 shall constitute failure of this test.

4.6.1.5 Performance.

4.6.1.5.1 Capacity. The FARE shall be assembled for operation as specified in the applicable technical manual. The FARE shall be operated for a period of nine continuous hours and flow rate shall be determined at the beginning of each hour by means of calibrated flow meters. Nonconformance to 3.13.1 shall constitute failure of this test.

4.6.1.5.2 Compatability. With the FARE assembled and operating as specified in 4.6.1.5.1, an inspection of all components and fittings shall be made. Nonconformance to 3.13.2 shall constitute failure of this test.

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4.6.1.5.3 Continuity. With the FARE assembled and operating as specified in 4.6.1.5.1, an ohmmeter shall be used to determine electrical continuity of the static ground assemblies. Nonconformance to 3.13.3 shall constitute failure of this test.

4.6.1.6 Transportability. Tests shall be conducted in accordance with the Association of American Railroads (AAR) recommended procedures for conducting field impact tests of loaded freight cars. The FARE shall be blocked, braced, and tied down in accordance with procedures contained in the technical manuals or as specified in AAR Rules Governing the Loading of Department of Defense Materiel on open top cars. The FARE prepared for shipment will be loaded and tied down on an open top flatcar. Instrumention tests will be conducted at speeds of 4, 6, and 8 mph in one direction and at 8 mph in the opposite direction. After each impact the load and the tiedowns shall be inspected to determine the amount of load shaft, condition of blocking and bracing or evidence of possible failure. Once the test has begun, there shall be no readjustment of load, nor any reconditioning of bracing and chock material or tiedowns. After completion of the rail impact testing, an operational check-out of the FARE shall be conducted. Any failure of tiedown attachments or any damage or permanent deformation to the unit or failure of the FARE to operate as specified after rail impact testing shall constitute failure of this test.

4.6.1.7 Safety. The FARE shall be evaluated throughout testing as specified in 4.6.1. Nonconformance to 3.19 shall constitute failure of this test.

4.7 Inspection comparison. The Government may select FARE at any time during the contract production period and subject these FARE to the examination and tests specified in 4.6. The inspection will be performed by the Government, at a site selected by the Government, on FARE selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction model and initial production FARE. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison FARE to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison FARE and shall be cause for refusal by the Government to continue acceptance of production FARE until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on FARE previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all FARE accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

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4.8 Inspection of packaging.

4.8.1 First article pack inspection. The first article pack shall be examined for the defects specified in 4.8.2.3. Presence of one or more defects shall be cause for rejection.

4.8.2 Quality conformance inspection of pack.

4.8.2.1 Unit of product. For the purpose of inspection, all components that will comprise a completed pack prepared for shipment shall be considered a unit of product.

4.8.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.8.2.3 Examination. Samples selected in accordance with 4.8.2.2 shall be examined for the following defects. AQL shall be 1.0 percent defective.

118.	Pump assembly not preserved in accordance with the referenced document as specified for level A.
119.	Filter-separator and replacement filter elements not preserved in accordance with the referenced document as specified for level A and commercial.
120.	Discharge hoses not drained and open ends not closed or covered as specified for level A.
121.	Discharge hoses not coiled and secured in the frames as specified for level A.
122.	Nozzles and fittings not placed in the canvas container and secured as specified for level A.
123.	Ends of suction hoses and ground rods not cushioned as specified for level A.
124.	Suction hoses and ground rods not placed in the canvas container and secured as specified for level A.
125.	Consolidation not as specified for level A.
126.	Packing not as specified for level A, level B, or commercial.
127.	Marking missing, illegible, incorrect or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 First article pack. The contractor shall furnish a first article pack for examination within the time frame specified (see 6.2), to prove, prior to starting production packaging, that the applied preservation, packing, and marking comply with the packing requirements of this specification. Examination shall be as specified in section 4 and shall be subject to surveillance and approval by the Government (see 6.4). The first article pack may be accomplished utilizing either the first article model FARE or a production FARE. If the first article model FARE is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government, when requested by the Government to facilitate comparison between the first article model FARE and production FARE.

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5.2 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.2.1 Level A.

5.2.1.1 Pump assembly. Components of the pump assembly shall be preserved in accordance with level A requirements of MIL-P-52746.

5.2.1.2 Filter-separator and replacement filter elements. The filter-separator and replacement filter elements shall be preserved in accordance with MIL-F-52556.

5.2.1.3 Discharge hose and fitting kits. Discharge hoses shall be drained of fuel or water and shall be coiled flat. Open ends shall be closed with appropriate caps or plugs or shall be covered with material conforming to L-P-378, type I, class 1, thickness 4 mil, secured in place with tape conforming to PPP-T-60, type IV. The coiled hoses shall be secured in the frames with the web straps provided. Nozzles and fittings shall be placed in the canvas containers in their designated places and shall be secured by the canvas container flaps and web straps provided.

5.2.1.4 Suction hose and ground rod kits. The suction hoses and ground rods shall be placed in the canvas containers in their designated spaces. The ends of each suction hose and ground rod shall be wrapped with cushioning material to prevent damage to the canvas container. The container flaps and web straps shall be secured.

5.2.1.5 Repair parts. The preservative application criteria and applicable methods of preservation of MIL-P-116 shall be used to preserve the repair parts.

5.2.1.6 Maintenance tools. Maintenance tools shall be preserved in accordance with PPP-P-40, level A.

5.2.1.7 Technical publications. Technical publications shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3.

5.2.1.8 Consolidation. The replacement filter elements, repair parts, maintenance tools and technical publications for each FARE preserved as specified shall be consolidated in a close-fitting box conforming to PPP-B-601, domestic type, style I, grade B. Closure shall be in accordance with the box specification. Strapping is not required. Box configuration shall be such as to permit the packing shown on figure 3.

5.2.2 Commercial. The filter separator and replacement filter elements shall be preserved in accordance with the requirements of MIL-F-52556. All other components of the FARE shall be preserved in accordance with ASTM D 3951.

5.3 Packing. Packing shall be level A, level B, or commercial as specified (see 6.2).

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5.3.1 Level A or level B. Packing for level A or level B shall be in accordance with table II and figures 2 and 3 for the level specified. The container shall be fabricated in accordance with PPP-B-601, table I and figures 2, 3, 4, and 5.

5.3.2 Commercial. The components comprising a complete FARE, preserved as specified in 5.2, shall be packed in a container in accordance with ASTM D 3951.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection (level A or level B) shall be in accordance with MIL-STD-129.

5.4.2 Commercial. Marking for commercial packaging shall be in accordance with ASTM D 3951. In addition the shipping container shall indicate the gross weight and cube. The packaging shall be labeled "FORWARD AREA REFUELING EQUIPMENT" in 2 inch letters on the sides and top of the box.

6. NOTES

6.1 Intended use. The FARE is intended for refueling helicopters in forward operational areas and for refueling military vehicles and powered equipment. The FARE shall be used in all environmental conditions listed herein.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame required for submission of first article model (see 3.2).
c. Color required when other than as specified (see 3.15).
d. When the welding process will not be at the option of the contractor (see 3.17.4).
e. When a first article sample is required (see 4.3).
f. Time frame required for submission of the first article pack (see 5.1).
g. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the items shall be preproduction models. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the document's first article.

6.4 First article pack. Any changes or deviations of production pack from the approved first article pack will be subject to the approval of the contracting officer. Approval of the first article pack will not relieve the contractor of his obligation to preserve, pack, and mark the FARE in accordance with this specification.

6.5 Lubricants. MIL-STD-838, Lubrication of Military Equipment, prescribes the policy for using specification-type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113, Guide for the Selection of Lubricants,

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Fluids, Preservatives and Specialty Products for Use in Ground Equipment Systems. The contracting officer should note that unless otherwise authorized by the US Army Belvoir Research, Development, and Engineering Center (ATTN: STRBE-VF), lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under chapter 2 of MIL-HDBK-113.

6.6 Quality assurance provisions (QAP). The contracting officer should require the contractor to maintain records of all QAP inspections. A suggested paragraph is as follows:

"The contractor shall maintain complete records of all examinations and tests performed to verify the requirements of classified QAP characteristics. The records shall include, as a minimum, lot size, sample size, drawing requirements, actual measurements, number and type of deficiencies found, quantity approved, quantity rejected, and corrective action taken when applicable."

6.7 Definition.

6.7.1 Quality assurance provisions (QAP). A QAP is a contractual requirement that supplements section 4 of the specification. QAP indicate the minimum requirements which must be inspected on the product drawings to verify the design objectives of the product and assure interchangeability of repair parts.

6.8 Subject term (key word) listing.

Aircraft refueling
Fuel-dispensing system
Ground vehicle refueling
Powered equipment refueling
Refueling system

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

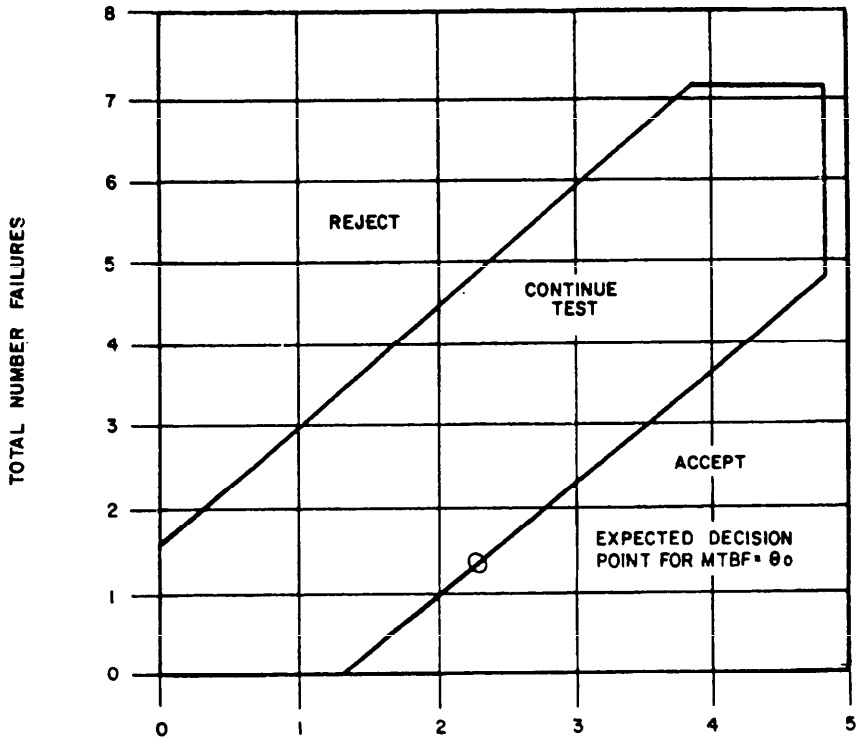
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DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 1. Accept-reject criteria.

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NOTES:
1. TOP TO BE INVERTED AS SHOWN.
2. LENGTH, WIDTH AND HEIGHT DIMENSION TO BE DETERMINED AS REQUIRED TO PLACE AND SECURE COMPONENTS ON THE SKID BASE AS SHOWN ON FIGURE 3; WITH ONE INCH CLEARENCE BETWEEN ALL COMPONENTS MOUNTED ON THE SKID BASE AND ONE INCH CLEARENCE BETWEEN THE COMPONENTS AND THE CONTAINER ENDS, SIDES AND TOP.
3. SEE TABLE II FOR LIST OF MATERIALS.
4. NAILING SHALL BE IN ACCORDANCE WITH PPP-B-601.

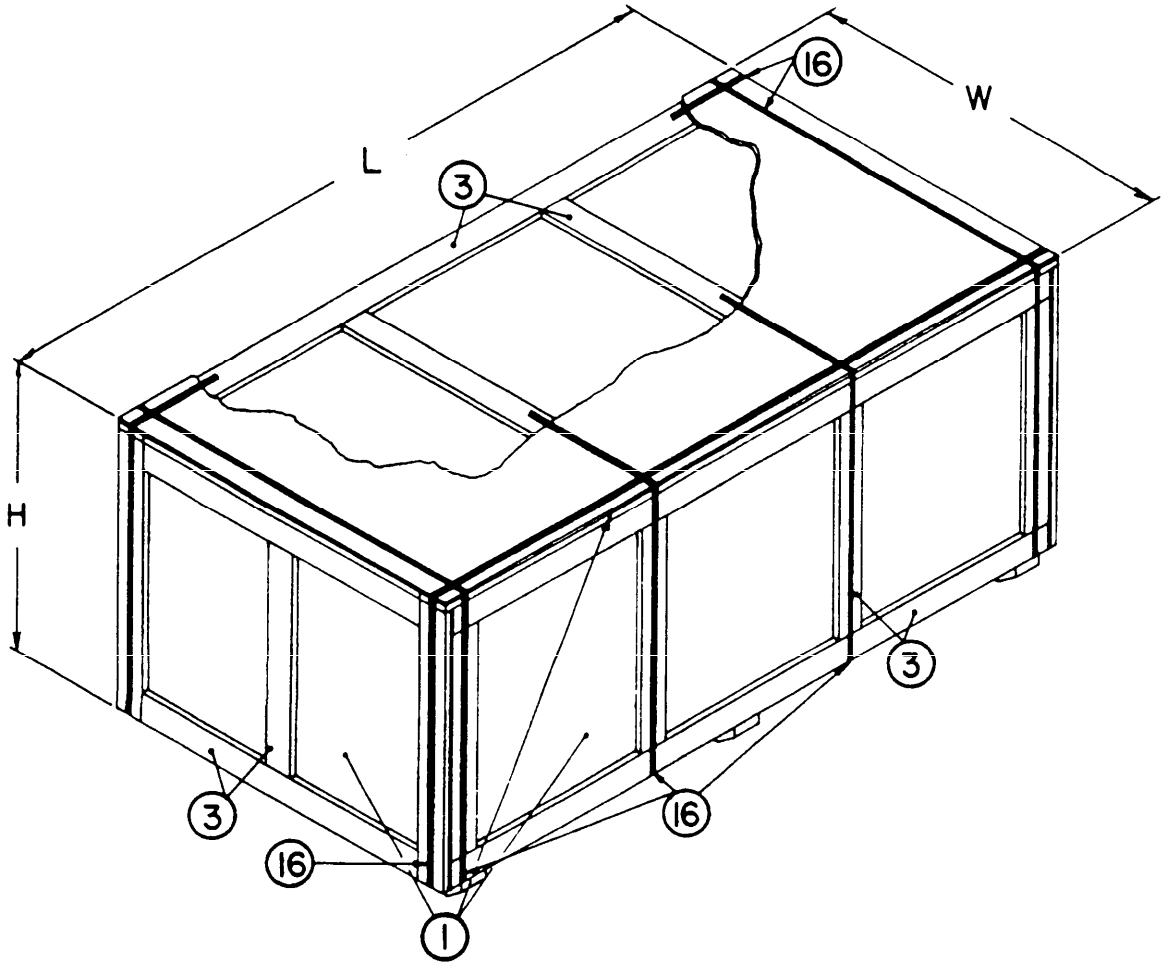
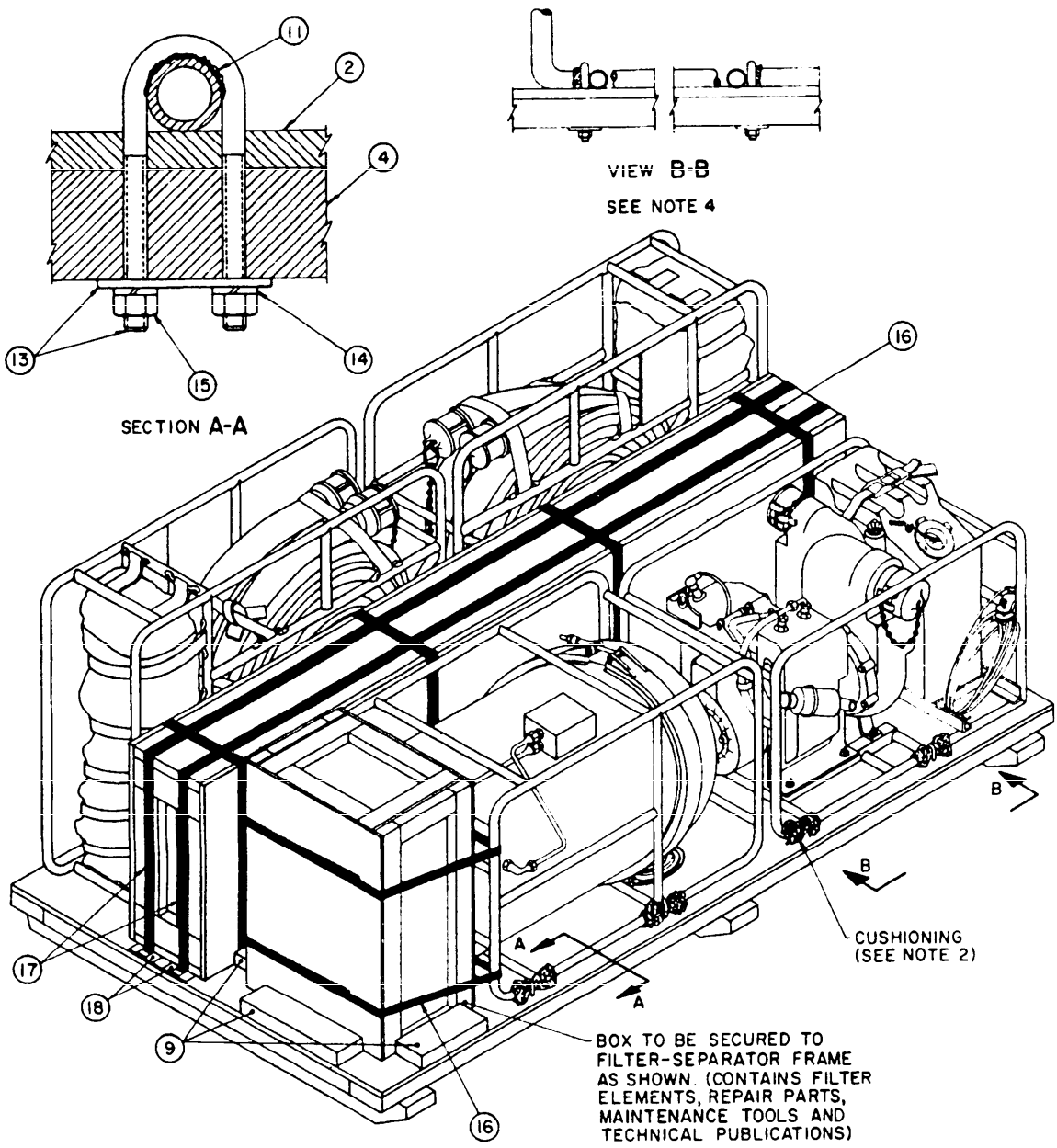


FIGURE 2. Cover details.

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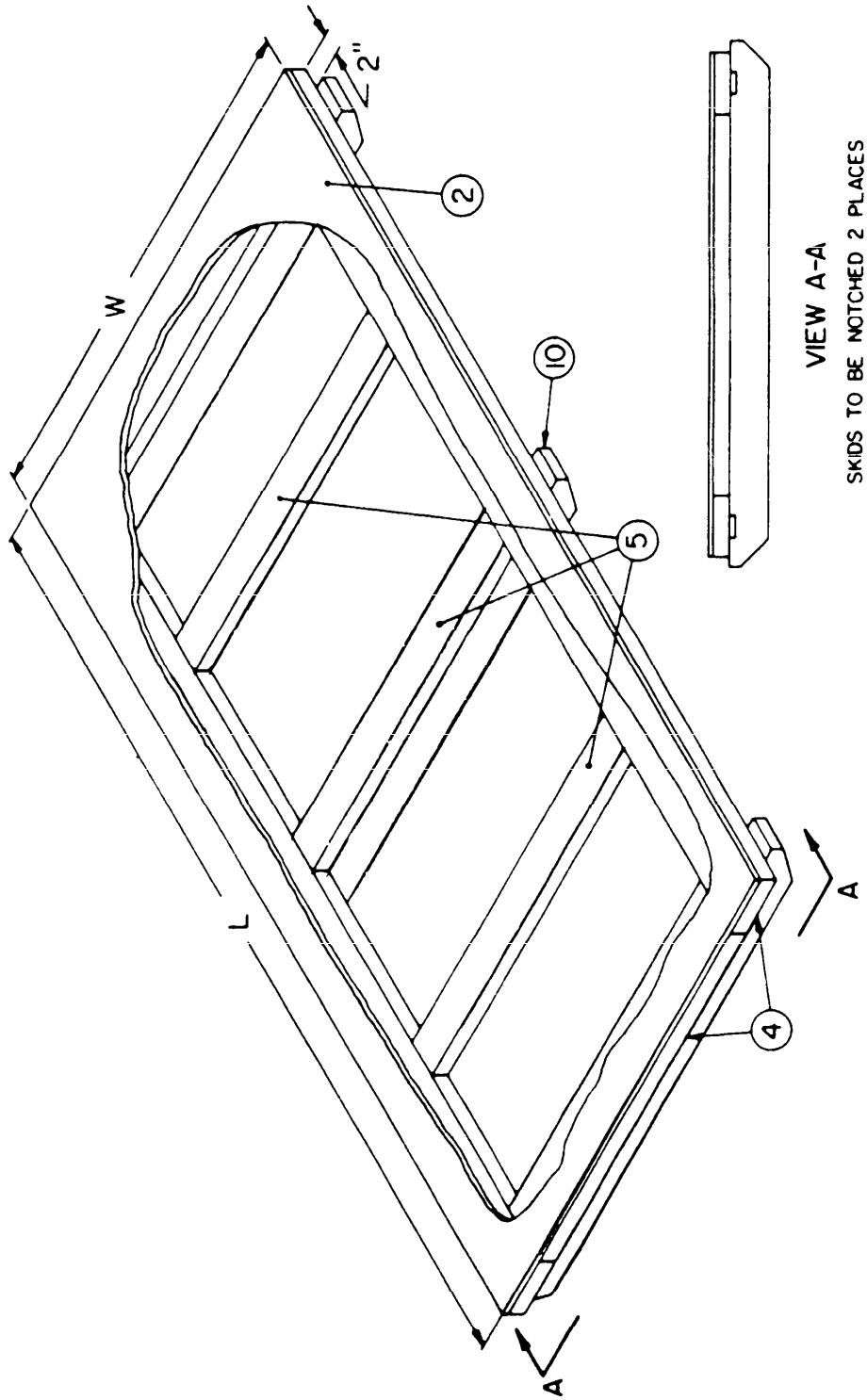
NOTES:

1. IN CENTER AREAS WHERE U-BOLTS WILL NOT SECURE THROUGH A CLEAT MEMBER OF THE BASE, SLEEPERS OF NOMINAL 2 INCH THICK LUMBER SHALL BE PROVIDED.
2. CUSHIONING, FIND NO. 11, SHALL BE PROVIDED UNDER ALL U-BOLTS AS SHOWN.
3. GROUND WIRES SHALL BE SECURED WITH TAPE, FIND NO. 12.
4. U-BOLTS SHALL BE POSITIONED ON ALL FRAME MEMBERS AS SHOWN TO RESIST MOVEMENT IN EITHER DIRECTION.

FIGURE 3. Packing arrangement and details.

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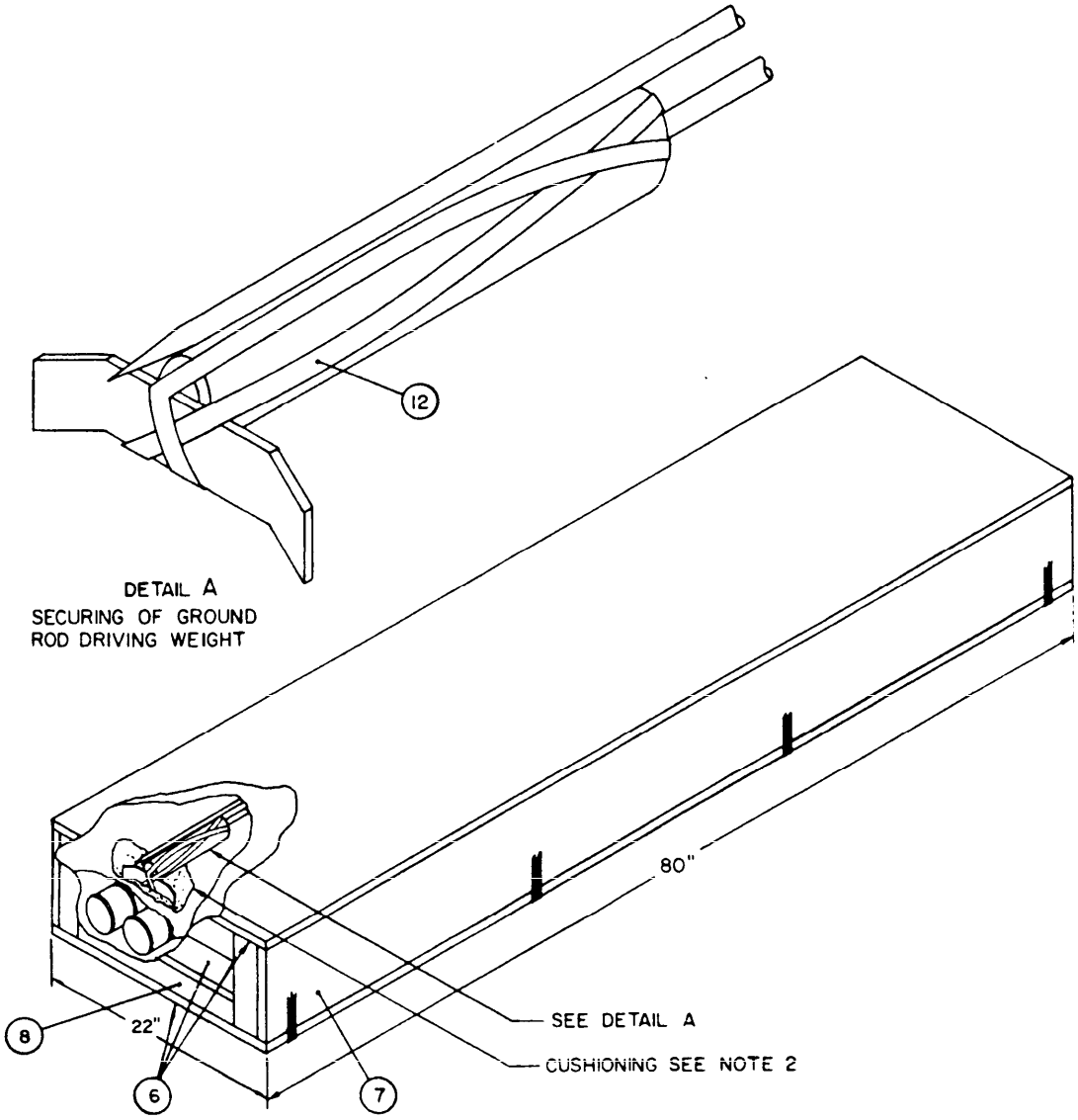


NOTES
1 LENGTH AND WIDTH DIMENSION TO BE DETERMINED AS REQUIRED FOR MOUNTING COMPONENTS (SEE NOTE 2 FIGURE 2).
2 SEE TABLE II FOR LIST OF MATERIALS

FIGURE 4. Skid details

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NOTES:

1. CUSHIONING, FIND NO. 11, SHALL BE PROVIDED BETWEEN BOX ENDS AND CONTENTS (SUCTION HOSES AND GROUND RODS).
2. ENDS OF GROUND RODS SHALL BE CUSHIONED, WITH FIND NO. 11, BEFORE PLACING THE GROUND RODS IN THE CANVAS CONTAINER. CUSHIONING SHALL BE SECURED WITH TAPE, FIND NO. 12.
3. SEE TABLE II FOR LIST OF MATERIALS.

FIGURE 5. Box details.

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TABLE I.

Preservation, packaging and packing material

FIND NO	IDENTIFICATION	DESCRIPTION	LEVEL	
			A	B
1	NN-P-530, PANEL, ENDS, SIDES, AND TOP (NOTE 1)	PLYWOOD, 1/4 THICKNESS	AR	
1	NN-P-530, PANEL, ENDS, SIDES, AND TOP (NOTE 2)	PLYWOOD, 1/4 THICKNESS		AR
2	NN-P-530, PANEL, SKID BASE (NOTE 1)	PLYWOOD, 1/2 THICKNESS	AR	
2	NN-P-530, PANEL, SKID BASE (NOTE 2)	PLYWOOD, 1/2 THICKNESS		AR
3	CLEATS, TOP PANEL, END PANELS AND SIDE PANELS	LUMBER, SOFTWOOD, 1X4 NOMINAL	AR	AR
4	CLEATS, SKID BASE, ENDS AND SIDES	LUMBER, SOFTWOOD, 2X6 NOMINAL	AR	AR
5	CLEATS, SKID BASE, INTERMEDIATE	LUMBER, SOFTWOOD, 2X4 NOMINAL	AR	AR
6	NN-P-530 PANEL, TOP, BOTTOM, AND END, (NOTE 1)	PLYWOOD, 3/8 THICKNESS	AR	
6	NN-P-530 PANEL, TOP, BOTTOM, AND END, (NOTE 2)	PLYWOOD, 3/8 THICKNESS		AR
7	PANEL, SIDES	LUMBER, SOFTWOOD, 1X10 NOMINAL	2	2
8	CLEATS, ENDS	LUMBER, SOFTWOOD, 1X4 NOMINAL	AR	AR
9	BLOCKING, CONSOLIDATED CONTAINER	LUMBER, SOFTWOOD, 1 THICK NOMINAL	AR	AR
10	SKID	LUMBER, SOFTWOOD, 4X4 NOMINAL	AR	AR
11	PPP-C-843, TYPE II, CLASS A	CUSHIONING	AR	AR
12	PPP-T-97, TYPE IV	TAPE, PRESSURE SENSITIVE ADHESIVE FILAMENT REINFORCED	AR	AR
13	U-BOLT WITH PLATE, 5/16-18UNC X 1 PIPE SIZE 3-9/16 INSIDE LG	STL, CAD OR ZINC PL	6	16
14	MS35338-45, 5/16 NOMINAL SIZE	WASHER, LOCK, SPRING, HELICAL (MED SERIES) CS, CAD PL	32	32
15	MS35649-2312, 5/16-18UNC-2B	NUT, PLAIN, HEX, MACH SCREW STL, CAD PL	32	32
16	QQ-S-781, TYPE I, FINISH B	STEEL, STRAPPING FLAT, .020 X 3/4 W	AR	
16	QQ-S-781, TYPE I, FINISH A	STEEL, STRAPPING FLAT, 020 X 3/4 W		AR
17	QQ-S-781, TYPE I, FINISH B	STEEL, STRAPPING FLAT, 035 X 1-1/4 W	AR	
17	QQ-S-781, TYPE I, FINISH A	STEEL, STRAPPING FLAT, 035 X 1-1/4 W		AR
18	ANCHOR PLATE	STEEL	4	4

NOTES

1. GROUP A CONFORMING TO ANSI/HPMA 1983, TYPE I, GRADE 3-4 OR GROUP B CONFORMING TO PS-1, GRADE C-D, WITH EXTERIOR GLUE

2. GROUP A CONFORMING TO ANSI/HPMA 1983, TYPE I, GRADE 3-4 OR GROUP B CONFORMING TO PS-1, GRADE C-D, WITH EXTERIOR GLUE

X-2737B

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-F-52749B(ME)	Forward Area Refueling Equipment (FARE)

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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